SolariNet — Directory Services & SSO

A cross-platform identity backbone for the homelab: Active Directory, Kerberos, LDAP, and web SSO — built on `radium` · 2026-07-05

> **What went live.** `radium` (a Pi CM5, reinstalled to Debian 13 on NVMe) is now a **Samba Active Directory domain controller** for `AKORIA.ORG`, with **Keycloak** layered on top for web/OIDC single-sign-on federated to AD, a first fleet host (`tungsten`) joined and proven end-to-end, an NFS home export live on `cesium`, three new Ansible roles, and a complete (undeployed) OIDC integration for the SolariNet dashboard.

Architecture

One directory, every platform. Samba AD DC is the authoritative identity store (Kerberos + LDAP + DNS). Linux fleet hosts join it via SSSD/realmd; macOS/Windows can join natively; websites federate through Keycloak (OIDC/SAML). Home directories live once on `cesium` and follow the user to any host via autofs+NFS. UIDs are SID-algorithmic, so a user is the same numeric identity fleet-wide.

```
mac / windows / linux / ssh ──►  Samba AD DC (radium.akoria.org)  ◄── SSSD/realmd (fleet
                                   ├─ Kerberos KDC (:88)   ├─ LDAP/LDAPS (:389/:636)
websites / dashboard ──OIDC──►  Keycloak (sso.akoria.org:8443) ──LDAPS──► AD federation
any joined host ──login──►  autofs /home/akoria/<user>  ──NFS──►  cesium:/export/home/<u
```

What's live

COMPONENT	WHERE	STATE
Samba AD DC — `AKORIA.ORG`	radium 10.1.0.10	live · Kerberos/LDAP/DNS/SMB verified

Keycloak 26.6.4 (PostgreSQL, HTTPS :8443)	radium · sso.akoria.org	live · realm `akoria`, AD federated over LDAPS
OIDC client `solarinet-dashboard`	Keycloak akoria realm	created · AD login → token verified
Fleet join proof	tungsten 10.5.0.5	joined · SSO login + auto-home + TGT verified
NFS home export	cesium `/export/home`	live · root_squash, crossmnt
Ansible roles: directory-join / shared-home / dev-tools	deploy/fleet/ansible	authored + validated (check-mode) · uncommitted
Dashboard OIDC login	dashboard/api + public	code done + verified · not deployed/committed

Credentials (AD Administrator, Keycloak admin, DB, dashboard client secret, AD user `jason`) are stored in agent memory and in `/root/.kc-*` on radium — deliberately kept out of this committed doc.

Ansible (deploy/fleet/ansible)

Three opt-in roles were added alongside the existing base-os/common-stack/solarinet-client, wired into `site.yml` and gated by `solari_ad_enabled` / `solari_home_enabled` / `solari_devtools_enabled`, with the AD contract in `group_vars/all.yml`:

- **directory-join** — install SSSD/realmd stack, point DNS at the DC, idempotent `realm join`, declarative `sssd.conf` (SID id-mapping for stable UIDs), mkhomedir, AD-group sudo. Join password is supplied at run time via vault/`--extra-vars` and never stored.
- **shared-home** — autofs + NFS client; wildcard map mounts `/home/akoria/<user>` from cesium on demand.
- **dev-tools** — the CLI toolset (zsh, jq, cloc, bat, ripgrep, eza, fd, cloc, starship, thefuck…), system-wide oh-my-zsh + powerlevel10k, and the fleet-adapted dotfiles vendored from xenon (deployed via `/etc/skel` so new directory-user homes inherit them).

```
# run a role against a host (from benzene or xenon):
ansible-playbook -i inventory playbooks/site.yml --limit <host> --tags ad \
  -e solari_ad_enabled=true -e solari_ad_join_password='<AD admin pw>'
```

> **Dashboard OIDC — finalize before it goes live.** The code is complete, additive, and keeps local login untouched, but is intentionally *not deployed or committed*. To turn it on:
>
> 1. Add the dashboard's public callback (`https://<dashboard-host>/api/auth/oidc/callback`) as a Valid Redirect URI on the `solarinet-dashboard` Keycloak client; add a *Group Membership* mapper (claim `groups`) so `domain admins` maps to the admin role.
> 2. In `solari-auth.json` → `oidc` : set `enabled:true` , `redirectUri` , and `caBundle` to a PEM trusting radium's cert (`openssl s_client -connect sso.akoria.org:8443 … > radium-ca.pem`).
> 3. Ensure the dashboard host resolves/reaches `sso.akoria.org:8443` ; deploy; browser-test the "Sign in with Akoria SSO" button.

Deferred (your call)

- Commit the Ansible roles + dashboard OIDC work (currently in the working tree).
- Roll out dev-tools/shared-home to real hosts; live end-to-end shared-home test (login → NFS home).
- Join production hosts — xenon (⚠ the box the agent runs on; keep a fallback login), cesium, benzene.
- DNS reach for non-fleet mac/Windows clients so they can find the DC.
- Replace Keycloak's self-signed cert + Samba's autogenerated TLS cert with a proper CA/SAN cert.
- Kerberized NFS (sec=krb5p) to replace sec=sys for the home export.